DEPARTMENT OF COMPUTER SCIENCE (BS IT PROGRAM)
OBJECT ORIENTED PROGRAMMING LAB(CSL-210)
PROJECT PROPOSAL FORM

SEMESTER Name	Term Name	BS (IT)	Lab Engineer: Madiha Aslam

TITLE OF PROJECT

Password Manager and Analyzer

Group Members

S#	Student(s) Name	Registration Number
1	Daniyal Afzaal	94821
email:	02-135251-034@student.bahria.edu.pk	No:
2	Shah Raiz Asghar	98848
email:	02-135251-022@student.bahria.edu.pk	No:
3	Muhammad Hasan	98850
email:	02-135251-040@student.bahria.edu.pk	No:

EXECUTIVE SUMMARY OF PROJECT PROPOSAL

This project is an OOP-based system that helps users securely store and evaluate their passwords through a clean, user-friendly **GUI interface**. Users will sign up, log in, and then manage multiple passwords for different platforms. All saved passwords will be **encrypted**, ensuring privacy and security. The system will also provide a **Password Analyzer** that checks password strength based on defined rules.
The project includes full data validation, proper file handling, and a structured OOP design

PROJECT PURPOSE, SCOPE AND OBJECTIVES

Purpose:
To build a secure, offline password-management tool that helps users save passwords safely and analyze their strength.

Scope
Includes:

- User sign-up & login
- Encrypted password storage
- Password strength analysis
- Data validation and file handling
- GUI for a user friendly interaction
- OOP-structured classes

Excludes:

- Cloud storage
- Multi-device syncing
- Use of external encryption libraries

Objectives:

- Provide a simple, secure system for managing passwords
- Implement analysis to guide users toward stronger password choices
- Demonstrate OOP principles through a real-world use case

Intended Users
Any user who wants a lightweight, offline password tool.

Value
Unlike generic apps, this project focuses on academic OOP implementation while still solving a practical problem

PROJECT DESCRIPTION (most important part)
The system consists of two main modules: **A. Password Manager** • User registration & login • Add, view, update, and delete saved passwords • Encryption before saving data • File-based storage • Simple search using linear traversal **B. Password Analyzer** • Check password strength (length, digits, symbols, uppercase/lowercase) • Score the password as Weak/Medium/Strong • Give improvement suggestions • Fully coded rule-checking methods **Implementation Approach** • Separate classes for User, PasswordEntry, Analyzer, FileManager, EncryptionHelper • Full use of OOP concepts (encapsulation, inheritance, abstraction, polymorphism) • File handling for persistent data • GUI will be implemented using Java Libraries • Custom-built encryption method
Course Concepts to be used: • Classes and Objects • Encapsulation • Inheritance & polymorphism • Abstraction • Exception handling • File handling • Lists/arrays • Searching • Data Validation • GUI Programming • Basic encryption logic

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TEAM PROFILE
We all are collectively skilled in programming, system design, documentation, testing, and presentation. All team members will contribute to coding, writing, reviewing, and presenting the project. Each member brings strengths in logic-building, organization, debugging, and communication, ensuring balanced teamwork and shared responsibility.

ASSUMPTIONS AND CONSTRAINTS

Assumptions

- Users operate the system on a single machine
- No internet or external storage is required
- Files will hold sufficient storage for all data

Constraints

- Limited time and academic resources
- Only file handling
- GUI will be simple
- It will follow OOP principles strictly

PROJECT DELIVERABLES NOT CHANGEABLE

Project deliverables will include:

- Software project proposal
- Final application
- Final project Report

For Teacher Use Only

REMARKS

Course Teacher Name		**Signature**		**Date**	
Lab Engineer Name		**Signature**		**Date**	